21 June 1972
SUPERSEDING
MIL-P-40096B
3 October 1967

MILITARY SPECIFICATION

POLISHER, PORTABLE, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of portable, hand-held, electric polisher with a carrying case.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant of Communications, Electronic and Associated Electrical Equipment

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-1337 - General Suppression System Design Requirements for Portable Electric Hand Tools.

FSC 5130

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standard Institute (ANSI)

B3.5 - Tolerances for Ball and Roller Bearings
B3.6 - Boundary Dimensions for Ball and Roller Bearings
B3.8 - Mounting Ball and Roller Bearings, Specifications For

(Application for copies should be addressed to the American Standards Institute, Inc., 1430 Broadway, New York, N. Y. 10018.)

National Bureau of Standards (NBS)

Handbook H-28 - Screw Threads Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Underwriters' Laboratories, Inc. (UL)

UL 45 - Standard for Electric Tools
UL 498 - Attachment Plugs and Receptacle

Electrical Construction Materials List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; Walt Whitman Road, Melville, N. Y. 11749; or 1655 Scott Blvd., Santa Clara, California 95050.)

National Fire Protection Association (NFPA)

NFPA No. 70 - National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

3. REQUIREMENTS

* 3.1 First article.- When specified, the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2, and 6.3).

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3.2 Standard product.- The polishers covered by this specification shall be a regular commercial product of the manufacturer, except for any changes necessary to comply with the specification requirements.

* 3.3 Electrical requirements.-

* 3.3.1 Conformance to standards of Underwriters' Laboratories, Inc..- The polisher shall conform to the requirements of UL 45 and UL 498 for electrical characteristics.

* 3.3.2 Certification.- Prior to approval of the first article, or if none is submitted, prior to approval of the first shipment, the supplier shall furnish satisfactory evidence that the polisher he proposes to furnish under this specification meet the requirements specified in 3.3.1. Acceptable evidence of meeting the requirements of UL 45 and UL 498 shall be the UL label, or listing with reexamination of UL. In lieu of the label, or listing with reexamination, the supplier may submit independent proof, satisfactory to the contracting officer or his authorized representative, that the polisher conforms to the applicable requirements of the published standards, including methods of tests of the UL Standards noted above. Compliance with UL requirements of UL regarding fire and casualty hazards does not absolve the supplier from complete requirements with the requirements of this specification in order to secure the acceptance of his polisher. In event of difference between any of the requirements of this specification and those of UL, this specification shall govern.

3.4 Materials.- The materials used shall conform to the applicable specifications and requirements specified hereinafter. When materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice.

* 3.5 Design and construction.- The polisher shall be a hand-held, electric motor-driven unit, designed to impart a rotary motion to a circular polishing bonnet mounted on a flexible pad attached to a spindle driven by the motor. The polisher shall conform to the mechanical and electrical characteristics specified in table I. The polisher shall have an electric motor, housing, gear train, bonnet, flexible mounting pad for rear bonnet, rear handle, tool rest, removable front handle, electrical components and a carrying case.

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* TABLE I.- Polisher characteristics

Duty	Nominal size	Pad (dia.) size	Full-load power output (min) 1/	No-load spindle speed (min)	Rated current (min)	Length without pad (max)	Weight without pad or cable (max)
Inter-mittent	7 inches	7 inches	1/2 HP	1700 RPM	5.0 amperes	17 inches	13 pounds

1/ Full load power shall be determined at minimum rated current.

* 3.5.1 Motor.- The motor shall be inclosed and mounted within the polisher housing. The motor shall be of the intermittent duty type and shall conform to the electrical characteristics specified in table I, and applicable requirements of UL 45. The motor shall operate on 120 volt, 60 hertz, single-phase alternating current or 115 volt direct current power source.

3.5.2 Power supply cable.- Each polisher shall be furnished with a cable and plug assembly as an integral part of the polisher. The cable shall be not less than 15 feet long, consisting of a three wire electrical cord, type S, SO, SJ, ST, SJO, or SJT in accordance with National Electrical Code NFPA 70. The cable shall terminate in a three-way, integrally molded connection plug conforming to UL 498 requirements for two pole, three wire grounding devices, rated at 15 amperes, and 125 volts.

* 3.5.3 Bearings.- The armature or rotor shaft shall be equipped with ball or roller bearings. The intermediate shaft and driving shaft shall be equipped with ball, needle, or roller bearings. Needle bearings shall be used only when the hardness of the supported shaft, size, thrust bearings and seals are adequate. Thrust bearings shall take the end thrust of the polisher spindle. Ball and roller bearings shall conform to ANSI Standards B3.5, B3.6 and B3.8.

* 3.5.4 Switch.- Switches shall conform to the safety requirements of UL 45 and shall be conveniently located for turning the current on and off. The switches shall be of the quick break type and shall be either an automatic release type or an "on" and "off" type. Switches that release automatically shall have a lock for holding the switch in the "on" position. "On" and "off" type switches shall be plainly marked to show the "on" and "off" positions.

* 3.5.5 Housing.- The housing shall be cast aluminum alloy or plastic, or a combination of both in accordance with the manufacturer's current commerical practice. Lubrication fittings or receptacles shall be provided for all bearings (except sealed bearings) and other moving or rotating parts.

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3.5.6 Handles.- The rear handle of the polisher shall be fastened to the rear of the housing. The front handle shall be removable and shall be capable of attaching to either side of the housing for either left or right hand use.

* 3.5.7 Fastening devices.- All screws, pins, bolts, and similar parts shall be installed with lock washers, lock nuts, or similar devices to prevent loosening or misadjustment due to vibration. Any fastening device required to be loosened, tightened or removed for repair, maintenance or adjustment shall not be swaged, peened, staked, or otherwise deformed to prevent the loosening of the device.

* 3.5.8 Gears.- Gears for driving parallel shafts shall be spur or helical. Gears for driving right angle shafts shall be bevel or spiral bevel. All gears shall run. Gears with straight teeth shall have the involute curve outline in a lubricant confined in a casing that shall not leak.

* 3.5.9 Backing pad.- The polisher shall have a 7-inch semi-flexible pad backing for the polishing bonnet in accordance with the manufacturer's standard practice. The pad shall be provided with a threaded or tapped adapter for attaching to the polisher drive shaft.

3.5.10 Bonnet.- The polishing bonnet shall be fabricated from either lamb's wool, or wool yarn sewed on to a canvas backing pad, in accordance with manufacturer's standard practice. The polishing bonnet shall be 7-1/2 inches in diameter and shall be provided with a flexible edging to fit snugly over the pad edge.

* 3.5.11 Tool rest.- A saddle or tool rest shall be fastened to the top of the motor housing for resting the polisher with the bonnet disc facing up.

3.5.12 Threads.- All threaded parts shall comply with the requirements of National Bureau of Standards Handbook H-28.

* 3.5.13 Carrying case.- A metal carrying case in accordance with the manufacturer's standard commercial practice shall be furnished. The case shall have a hinged cover and a carrying handle.

* 3.6 Performance.- When tested as specified in 4.4.1, the polisher (1) shall operate at no-load spindle speed and full-load power output at rated current input as specified in table I with gears running smoothly and quietly; (2) shall have no visible evidence of lubricant leakage or loosening of components including motor parts, bearings, gears, and gear casing.

* 3.7 Electromagnetic compatibility.- When specified (see 6.2), the polisher shall meet the electromagnetic compatibility requirements of MIL-STD-1337 when tested as specified in 4.4.2.

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* 3.8 Fungus control.- The electric circuitry, including all components and connections, shall be protected from the effect of water and fungus in accordance with MIL-T-152 with exceptions as follows:

(a) Components or circuit elements that are inherently fungus and water resistant, or hermetically sealed.

(b) Components or circuit elements when function will be adversely affected by the treatment.

Varnish, when used, shall be applied to give a minimum dry thickness of 0.001 inch on component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

* 3.9 Identification plate.- A corrosion-resistant metal identification plate shall be provided, showing the manufacturer's name, serial number, maximum size or capacity of the polisher, voltage (AC or DC) current rating (amperes) and no load speed (revolutions per minute) in accordance with the manufacturer's standard commercial practice.

3.10 Expendable parts.- Each polisher shall be provided with three 7-1/2-inch diameter bonnets.

3.11 Finish.-

3.11.1 Polisher.- All exposed metal and working surfaces, except corrosion resisting parts, shall be finished in accordance with the manufacturer's standard practice.

3.11.2 Carrying case.- All surfaces of the carrying case shall be finished in accordance with manufacturer's standard practice.

3.12 Workmanship.- The polisher shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, deteriorations or malformations. There shall be no sharp edges, burrs, or slivers.

* 3.12.1 Finish coat.- The finish coat of the polisher and carrying case shall be continuous, smooth, adherent, without discoloration or foreign material imbedded, and contain no sags, runs, drips, creeps, laps, bubbles, cracks, pits, lumps, flux or orange peel. No rust, rough grinds or toolmarks shall show.

* 3.12.2 Welding, brazing and soldering.- Welding, brazing or soldering shall be free from flux, oxide, scale, paint, grease, and other foreign matter and shall be continuous, sound, smooth, and free from porosity, cracks, and incomplete fusion.

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3.12.3 Bolts, nuts, screws, studs and other threaded fasteners.- Threaded fasteners shall not be broken, stripped, fractured, or loose.

* 3.12.4 Electrical wiring and connections.- Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall not be tight and under strain. Conductor and terminal parts shall be mechanically secured before soldering.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a first article is required it shall be examined for defects in table II, specified dimensions, and tested as specified in 4.4.1 and 4.4.2.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- A lot shall consist of all polishers offered for inspection at one time. The sample unit shall be one completely fabricated polisher with carrying case.

4.3.2.1 Visual examination.- The completely assembled end item shall be examined for defects in finish, construction, workmanship and marking. Defects shall be as classified in table II. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not in accordance with manufacturer's standard practice finish coat not continuous, smooth, adherent, discolored, imbedded foreign matter; sags, runs, drips, creeps, lap, bubbles, cracks, pours, pits, lumps, flux or orange peel, rust, rough grinds or toolmarks showing thru coating.		X
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction (workmanship)	Component fractured, broken, punctured, torn, dented, creased, bowed, deteriorated, or malformed.	X	
	Component misplaced or not in proper alignment.	X	
	Any functioning component that requires abnormal force to operate.		X
	Component loose, not connected or joined properly.	X	
	Fastening device requiring loosening swaged, peened, staked or otherwise permanently fastened.	X	
	Provisions for lubrication not provided when required.	X	
	No removable covers on housing.	X	
	Carrying case handle loose, compartments not adequate, cover not hinged.	X	
Welding, brazing, soldering	Missing, incomplete, burn holes, pits, fractured and undercut; not fused, continuous, sound.	X	
	Smooth and uniform.	X	
	Not free of scale and flux deposits.		X
Bolts, nuts, screws, and other threaded fasteners	Missing, broken, stripped or fractured.	X	
	Threaded fastener loose.		X
	Lock washers, lock nuts, locking devices missing.		X

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* TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (workmanship) (cont'd)			
Electrical assembly	Wiring cut, abraded.	X	
	Wiring tight and strained insulation stripped	X	
Identification marking	Missing, incomplete, not legible.		X

4.3.2.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensional requirements. Any deviation from the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.2.3 Testing of end item.- When a first article is not required, the initial production unit shall be tested as specified in 4.4.1 and 4.4.2. Any noncompliance shall be cause for rejection.

4.3.3 Code and standards compliance.- Proof of compliance with the requirements of 3.3 shall be made available to the Government representative.

* 4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging and packing preservation and marking conform to PPP-P-40.

4.4 Tests.-

* 4.4.1 Performance test.- The polisher shall be tested for compliance with requirements of 3.6 as follows:

(1) The polisher shall be operated for 1 hour. The spindle speed and brake horsepower at the specified rated current input shall be measured and the running of gears checked to determine compliance with the requirements of 3.6 (1). Any nonconformance shall constitute a defect.

(2) At the completion of the above test, the polisher shall be examined to determine compliance with the requirements of 3.6 (2). Any nonconformance shall constitute a defect.

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* 4.4.2 Electromagnetic compatibility test.- The first article shall be tested as specified in MIL-STD-1337 to determine compliance with 3.6.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with PPP-P-40 (see 6.2).

6. NOTES

6.1 Intended use.- The intermittent duty, portable electric polisher covered by this specification is intended for use in polishing lacquered and painted surfaces on automobiles or furniture. The polisher can be used on flat or curved surfaces.

* 6.2 Ordering data.- Procurement documents should specify the following, as applicable:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.2 and 6.3).
(c) When electromagnetic compatibility is required (see 3.6).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1).

* 6.2.1 Contract data requirements.- Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

* 6.3 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:

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User activity:

Army - CE

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Bulletin, Inflatable, Elevator [illegible] MIL-F-[illegible]

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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